MILITARY SPECIFICATION

CRANE, CONTAINER HANDLING,
AIR TRANSPORTABLE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-propelled, diesel-engine-driven, wheel-mounted, 4 by 2, 360 degree (°) revolving superstructure, hydraulically operated, air transportable, ISO container handling crane.

2. APPLICABLE DOCUMENTS

2.1 *Government documents*.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3810

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MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-C-3580 - Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and Their Attachments, Packaging of.
MIL-A-8421 - Air Transportability Requirements, General Specification for.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Miliary Equipment.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B30-5 - Mobile and Locomotive Cranes.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117 - Salt Spray (Fog) Testing Standard, Method of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 668 - Series 1 Freight Containers - Classification, External Dimensions and Rating.

(Application for copies should be addressed to American National Standard Industry, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The crane shall be hydraulically operated, diesel-engine-powered, hydraulically propelled through two wheels, 360° full revolving superstructure, 14 ton capacity, and configured to be within the limits of the transport mode envelope for shipment by aircraft.

3.2 First article. When specified the contractor shall furnish a crane for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All cranes furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the crane.

3.6 Safety. The crane shall comply with applicable OSHA regulations and ANSI B30.5 except fenders are not required. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded, or insulated, to the extent necessary to eliminate the hazard. The principal platform walking surfaces shall be of an anti-skid type. Ladders, steps, and handholds shall be provided in such quantity and of such size on the sides of the crane cab or superstructure that entrance thereto and exit there from may be unhampered and nonhazardous. Engine cooling fans shall have heavy grille or ring type guards.

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3.7 Stress requirements. Design of the completely assembled crane shall be such that the maximum unit stress imposed will not exceed the minimum strength margins contained in SAE J1063. Tests, in accordance wth SAE J1063, shall have been performed on the model crane being provided and test results shall be made available for government inspection at the time of first article tests.

3.8 Air transportability. The crane shall be of a size and configuration so that it can be loaded into large cargo aircraft (C-130 and C-141) as a unit.

3.8.1 Dimensions. The crane shall present a low profile for shipment by aircraft. When prepared for shipment the following overall dimensions shall not be exceeded:

a. Height - 102 inches.
b. Width - 96 inches.
c. Length - 240 inches.

3.8.2 Aircraft floor loading. Axle loads, wheel loads, tire footprint loads, and general floor loads, as determined from the plan view of the equipment, shall conform to the fuselage zone and compartment limitations for the aircraft concerned.

3.8.3 Self-loading and discharging capability. The crane, under its own power, shall approach, negotiate, and depart a ramp 15 feet long at an angle of 17° which connects two horizontal surfaces without any part of the crane chassis touching any of the surfaces.

3.8.4 Tiedown criteria. Tiedown criteria shall be in accordance with 3.16.

3.8.5 Shipping weight and center-of-gravity location. The shipping weight of the crane, in an air transportable condition, shall be marked in a conspicuous location. The center-of-gravity along each axis influencing the method of loading and tiedown shall be marked on the crane.

3.8.6 Readiness. The crane shall be immediately available for operating after air landing. Disassembly to achieve loading into the aircraft is not permitted.

3.9 Performance. The crane shall be capable of independently and simultaneously elevating the boom and rotating the superstructure 360°. Superstructure slewing speed shall be not less than 2 revolutions per minute. The crane, without load, shall be capable of attaining and maintaining a travel speed, on level improved roads (see 6.5.1), of not less than 5 miles per hour (mph) and capable of traveling on unimproved roads (see 6.5.2); and shall be capable of negotiating 15 percent side slopes; and traveling up not less than a 30 percent grade on an improved road without engine stall. With the drive system disconnected, the crane shall be capable of being towed up to 25 mph on level paved roads.

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3.9.1 Lift capacity. The crane lift capacity shall be not less than 14 tons at a radius of 15 feet with the outriggers set.

3.9.2 Stability. With lubricating and cooling systems full, fuel tank(s) at least half full, with crane resting on firm, level, and uniform supporting surface, with outriggers unsupported, with the boom fully retracted and in travel position, the weight on the front axle shall be within 40 to 60 percent of the total weight of the crane; and shall conform to the following additional requirements:

a. With the longitudinal axis of the boom positioned 90° to the longitudinal axis of the carrier, the boom retracted and set at the maximum possible elevation, the total, load on all wheels on the side of the carrier under the boom shall not be less than 15 percent of the total weight of the crane.
b. With the longitudinal axis of the boom positioned in line with the longitudinal axis of the carrier, in either direction, the boom retracted and set at the maximum possible elevation, the total load on all wheels under the lighter loaded end of the carrier shall not be less than 15 percent of the total weight of the crane.

3.9.3 Power load lowering. Power load lowering by hydraulic control shall be provided for the winch to facilitate controlled load lowering or lifting at any boom lengths or elevation. The power down shall have sufficient capacity to raise, hold, and lower a load of not less than 7,000 pounds without slipping on a single part line. The power load lowering means shall provide positive and precise load control.

3.9.4 Rotation. The power transmission system used to rotate the superstructure shall be a hydraulic-mechanical arrangement with a continuous rotation of 360°. Swing speed shall be smooth and constant, and adequate control shall be provided to insure smooth starting and stopping. An automatic or manual type swing lock or brake, capable of locking the superstructure in any desired position shall be provided. The manual swing lock or brake controls shall be located within easy reach of the operator, while seated.

3.9.5 Winch capacity. The winch shall have sufficient capacity to make maximum rated lifts with shortest boom length and with the length of rope installed on the winch as specified in 3.10.4.

3.9.6 Rated load capacities chart. A permanent chart indicating all the rated load capacities on each boom reach and radii at 360° rotation, with outriggers set, shall be posted in the crane cab near the operator and also on the outside of the crane to be legible from ground level. The rated load capacity chart shall reflect the manufacturer's rating for the crane furnished, and not be limited to the minimum required capacities specified.

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3.9.7 Load clearance. With an ISO series 1CC freight container hoisted to any height, up to 70 inches above ground level and perpendicular to the boom centerline, the container shall not contact any part of the crane while slewing the superstructure 360° at 15 feet radius.

3.10 Crane. The crane assembly shall consist of a fully revolving superstructure with boom support, hydraulically operated telescoping type boom, a main winch with wire rope, hook block, cab type operator's station, engine controls, and necessary accessories.

3.10.1 Superstructure. The superstructure shall be mounted on the carrier and shall be designed to withstand the forces caused by lifting and swinging the maximum rated load. The carrier and outrigger mounting shall allow full 360° working range.

3.10.2 Boom. The boom shall be fabricated of alloy steel and shall be the hydraulically-operated telescoping type, consisting of a base section and sliding section(s). All extensions, retractions, and elevations of the boom shall be accomplished by hydraulic controls.

3.10.3 Hook block. The hook block assembly shall have adequate weight to insure payout of the cable without load on the hook when the hook is in any position. The hook shall be a forged steel swivel type with a design factor of 4.0 based on ultimate tensile strength, and shall be furnished with a safety latch on the hook. The hook shall not be painted. Block assembly shall be provided for attachment of hoist cable for not less than two-part purchase. Means shall be provided to secure hook block while traveling. The hook block furnished shall permit disassembly to visual inspection and nondestructive testing.

3.10.4 Rope. Load hoist rope(s) shall be extra improved plow steel, independent wire rope center with a design factor not less than 3.5. The manufacturer's standard length of wire rope shall be furnished.

3.10.5 Sheaves. All sheaves shall have a pitch diameter at least 18 times the rope diameter. Sheaves shall be adequately guarded to prevent the wire rope from leaving the sheave during operation. All sheaves shall be smooth and of the proper size for the wire rope furnished.

3.10.6 Hydraulic system. The hydraulic system shall include a pump(s), hydraulic rams, rods, full flow filters, reservoir, pressure relief valve, and all necessary lines, lockouts, restrictor, and control valves to insure positive control of boom hoist and rope hoist in all operations or loss of hydraulic power due to ruptured hoses. Hydraulic line circuit and filtration shall be in accordance with SAE J931. The pump(s) shall be driven by the crane engine and shall have sufficient capacity to operate all hydraulically powered components as specified herein. Means shall be provided in the hydraulic system for venting all air trapped in the system. All rods, which will be exposed during operation, shall have a hard chromium plating with crack-free thickness sufficient to withstand a 30-hour salt water spray test

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in accordance with ASTM B117. All high pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure, and a maximum operating pressure of at least two times the working pressure.

3.10.7 Radius indicating device. A dial or digital display of boom length and boom angle in feet shall be provided in the cab in view of the operator in his normal operating position. The system(s) shall comply with SAE J1180 and SAE J375.

3.10.8 Cab. A cab shall be mounted on the revolving superstructure and shall be fabricated from sheet steel, sheet aluminum, or fiberglass, having sufficient windows and glazed doors to permit at least 270° vision for the driver or operator in normal position. All glass shall be not less than 3/16-inch nominal thickness, safety type, and held in place by rubber or metal channels for easy replacement. Windows shall be arranged for easy opening and removal when required. All cab doors, whether of the sliding of swinging type, shall be adequately restrained from accidentally opening or closing when the crane is traveling or operating. All doors shall be provided with means for securely locking in the closed position. An adjustable and comfortable seat shall be provided in the cab and located to provide maximum unobstructed visibility for the operator. The cab shall be equipped with a diesel fueled or hot water heater, defroster, interior dome light, and air or electric operated windshield wiper. Electric or air horn, hand or foot operated, shall be provided. A positive swing lock for traveling shall be provided in the cab. The crane shall be provided with outside rearview mirrors on each side.

3.10.9 Controls. All carrier, and crane controls, lockouts, and indicators shall be located within easy reach of the operator while seated, and shall be readily accessible under all conditions of operations. All controls shall be clearly marked as to their use and function.

3.10.10 Engine. The engine furnished shall be of the diesel type having horsepower, torque, and speed characteristics to meet satisfactorily all the crane performance requirements specified herein. The engine shall be capable of meeting the performance requirements using diesel fuel conforming to VV-F-800. The diesel engine shall start and be ready for full load operation within 15 minutes in any ambient temperature from 125° Fahrenheit (F) to -20°F. When another priming system is required, it shall be of the measured shot type with storage capacity of at least 12 fluid ounces. The engine shall be furnished complete with at least the following accessories:

a. An instrument panel in the cab complete with a lubricating oil pressure gage, a fuel gage, a cooling liquid-temperature gage, a battery charge-discharge indicator, and a tachometer.
b. A fuel tank with sufficient capacity for 8 hours normal operation.
c. A 12-volt or 24-volt electric cranking system.
d. Battery(s) shall be shipped dry, charged, and sealed to prevent intrusion of atmospheric moisture.
e. A high capacity alternator.

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f. A cooling-liquid high temperature visible or audible safety device of the warning type.
g. A lubricating low oil pressure visible or audible safety device of the warning type.
h. An engine housing designed to permit easy access to facilitate maintenance of the engine.
i. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions indicated in this specification.

3.11 Carrier. Carrier frame shall be of the heavy steel fabrication type, with necessary braces and reinforcements to provide rigid support for the revolving superstructure, under maximum load, when the telescopic boom is extended at minimum through maximum radii at any position within 360° rotation. The carrier frame shall be mounted on four wheels with two wheels steerable and two-wheel drive. All wheels shall be equipped with interchangeable, high floatation pneumatic tires of compatible size for the load imposed. Power assisted full hydraulic steering shall be provided.

3.11.1 Drive system. The crane shall be equipped with a hydrostatic drive system for propelling the crane from 0 to 5 mph with the engine operating at the maximum governed speed. Propulsion speeds shall be infinite in both directions. In addition, the drive system shall be provided with a means to easily disconnect the drive to enable the crane to be towed at speeds up to 25 mph.

3.11.2 Brakes. The crane carrier shall be provided with the manufacturer's standard, foot controlled, service brakes. The service brakes shall be capable of bringing the crane to a complete safe stop in accordance with SAE J1152. A parking brake capable of holding the crane, on a longitudinal slope of not less than 15 percent without slippage, shall be provided.

3.11.3 Tires. Tires shall be tubeless type with traction type tread. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground. Tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be high flotation type.

3.11.4 Axle assembly. The front and rear axles shall be of sufficient capacity to sustain the imposed loads. The rear axle assembly shall be of the oscillating type, with a positive-type oscillation lockout device incorporated for air transportation. With lockouts in place, the rear axle assembly shall not oscillate more than 2 inches measured at centerline of tires. The lockout controls shall be manually operated.

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3.11.5 Outriggers. Four hydraulically operated outriggers, capable of extending 6 inches below ground level, shall be provided, one outrigger on each corner of the chassis. The outriggers shall be operated separately or in any combination from the operator's cab, and shall not collapse due to loss of hydraulic power. The outriggers' legs and pads shall be of a size to support the intended working loads without loss of crane stability or cause excessive ground pressure. When outriggers are fully retracted for transporting the crane, the maximum width shall be not greater than the transport mode envelope, see 3.8.

3.12 Lights. The following lights, as a minimum, shall be furnished and installed and shall be powered by the crane's electrical system:

a. Two sealed beam headlights (mounted on the carrier).
b. Two taillights.
c. Two stoplights.
d. Instrument panel light.
e. Two forward "looking" sealed beam worklights (mounted on the revolving superstructure).

3.13 Toolbox. The crane shall be provided with a metal toolbox. The toolbox shall have a hinged lid with a hasp suitable for padlocking. The toolbox shall be securely fastened to the crane in a readily accessible and protected location.

3.14 Slinging and tiedown provisions. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, except tying down attachments may be identified by stenciling or other suitable marking on the crane. Tying down marking shall clearly indicate that the attachments are intended for tying down of the crane on the carrier when shipped.

3.14.1 Slinging provisions. Permanently affixed slinging provisions that enable the crane to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.14.2 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the crane to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tie down provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209. In addition, tiedown provisions shall comply with the requirements of MIL-A-8412.

3.15 Identification plate. An identification data plate shall be furnished by the contracting officer, and permanently affixed in a conspicuous place on the crane. Nomenclature shall be "CRANE, CONTAINER HANDLING, AIR TRANSPORTABLE". Contractor shall stamp the necessary information and affix the plate.

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3.16 Instruction plates. The crane shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment.

3.17 Stenciling. The gross weight of the crane shall be stenciled on each side of the crane in such a matter as to be readily discernable to dock and air freight personnel. The recommended tire pressure shall be stenciled on the carrier by block or stencil type letters not more than 1-inch high.

3.18 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be green number 14064 conforming to FED-STD-595. Hazard stripes shall be applied to the rear of the carrier, boom point, and hook block cheek plates. Stripes shall be yellow number 13538 conforming to FED-STD-595, 2 inches wide, 2 inches apart and applied at a 45° angle.

3.20 Workmanship.

3.20.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.20.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.20.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.20.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 *Classification of inspections*. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each crane when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4.1 thru 4.4.10.2, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.11 and the packaging inspection of 4.5.

4.3 Examination. Each crane shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Crane load capacity test. The crane shall be completely serviced and subjected to the manufacturer's recommended break-in period. The load capacity test shall be performed on a level, paved surface by connecting a Baldwin SR 4 load cell or equal, between the lifting hook and a dead weight or the direct use of calibrated weights. The crane shall then lift the loads specified in the least stable direction. The crane, with load raised, shall be rotated 360° clockwise and counterclockwise. Actual tipping loads, when tipping load is the controlling factor, for each of the load conditions shall be determined in accordance with SAE J765. Eighty-five percent of each determined tipping load shall be compared with the applicable rated load.

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4.4.2 Speed tests. Speed tests shall be taken over a zero percent grade on an improved road. Time shall be recorded with a stopwatch accurate to 1 second. The average speed shall be recorded for the unit. Maximum average speed not less than that specifed in 3.9 when calculated by the following formula shall be attained:

$$V = \frac{0.682 \times D}{T}$$

Where: V = Speed in mph.
D = distance in feet.
T = Time in seconds.

A fifth wheel type recorder may be used.

4.4.3 Gradeability. The crane unit shall be tested for gradeability in compliance with 3.9. The crane unit, without load, shall negotiate a 30 percent grade on improved road without engine stall.

4.4.4 Power load lowering test. General inspection of the power load lowering system shall be made for adequate lubrication, ease of operation of controls, safety precautions, and for good manufacturing practices. The power load lowering system shall be tested as follows to determine conformance to 3.9.3:

a. Set outriggers, and position the revolving superstructure in the most stable craning position. Assemble a single line load equal to the maximum rated load divided by the recommended number of parts of hoist line reeving.
b. Position the boom angle for the minimum working radius.
c. Hoist the load 6 feet above ground level, ± 3 inches, or until interference with the crane boom is obvious.
d. Operate the power load lowering system in accordance with manufacturer's instructions, and lower the load under power.
e. When the load reaches the ground, immediately repeat operations (c) and (d). These operations shall be repeated until 20 cycles have been completed.

4.4.5 Radius indicating device test. The radius indicating device shall be tested to determine conformance to 3.10.7.

4.4.6 Road test. The crane shall be given a road test for a distance of not less than 1 mile at a speed of 5 mph over an unimproved road (see 6.5). In addition, to demonstrate steering ability the crane shall be operated forward and backward in two figure-8 patterns with hard right and left turns.

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4.4.7 Service brake test. Service brakes of the crane shall be tested for compliance with 3.11.2. Immediately after the crane comes to a stop, this test cycle shall be repeated for four additional cycles. All shoes, drums, and other brake components where there is indication of damage shall be disassembled and examined.

4.4.8 Parking brake test. The parking brake shall be tested for compliance with 3.11.2.

4.4.9 Slinging provisions.

4.4.9.1 Test procedure.

a. Lift the crane and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the crane. Determine the forward angles of application for each provision. Measure the clearance between each sling and the crane. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the crane by anchoring the main frame and subject each slinging provision to a force of 2-1/2 times the load it carried when suspended. This force shall be applied in the direction as determined in (a) above. An acceptable alternate method of test shall be to lift the crane as in (a) above and add weights to the crane until the force on each provision is 2-1/2 time the initial force. Hold the crane under either of the above loading conditions for not less than 90 seconds.

4.4.9.2 Failure criteria. Weld failure, permanent deformation of the crane or any slinging provisions, slings converging more than 24 feet above the lowest extremity of the crane, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.4.10 Tiedown provisions.

4.4.10.1 Test procedure. With the crane anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.4.10.2 Failure criteria. Weld failure, permanent deformation of the crane or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.4.11 Production unit operational test. Each production crane shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warmup period as recommended by the manufacturer. The crane shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operated promptly, fully, and without restriction or malfunction. Test for capacity loads specified in 4.4.1 shall be conducted.

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4.4.12 Production sample. Upon acceptance of the first production crane, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.5 Preparation for delivery inspection. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 4 of MIL-C-3580. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3580 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The crane is intended for handling 8x8x20 foot shipping containers at remote locations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When lubrication shall be other than as specified (see 3.18).
c. When a first article is required for inspection and approval (see 4.2.1 and 6.4).
d. When a preproduction pack is required and the time frame required for submission of the preproduction pack (see 4.5).
e. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29244(YD)	CRANE, CONTAINER HANDLING, AIR TRANSPORTABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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